

Numerical Simulations of Ice Melting: Investigating the Impact of Rotation and Salinity Using Dedalus 3

May 2024 - August 2024

Yasaman Yazdani

Abstract

Iceberg melting plays a pivotal role in influencing freshwater fluxes into the oceans, with significant implications for marine ecosystems and oceanic circulation. This study explores the combined effects of salinity and rotation on ice melting through two-dimensional numerical simulations using the Dedalus 3 framework. The simulations, grounded in modified phase-field models, incorporate the Coriolis force to assess how varying salinity levels and rotational velocities affect melting dynamics. My results indicate that initially, higher salinity accelerates melting due to freezing point depression, but it ultimately stabilizes thermal gradients, thereby reducing the melting rate. Additionally, higher salinity reduces the surface area of the remaining ice at the same volume, further decreasing melting rates. The influence of rotation becomes more pronounced at higher angular velocities and varies with the surrounding water's salinity. In freshwater, rotation slows the melting rate by limiting vertical convection. At 30 g/kg salinity, this reduction is observed initially but diminishes after stratification. Conversely, at higher salinities (100 g/kg and 200 g/kg), rotation initially slows melting, but once the system stabilizes, it enhances the rate, likely due to the onset of thermal wind balance.

Introduction

Iceberg melting is a critical process that contributes significantly to freshwater fluxes from ice sheets into the oceans, accounting for a substantial portion of freshwater loss in regions such as Antarctica and Greenland [1, 2]. The meltwater released by icebergs plays a vital role in marine ecosystems, influencing nutrient availability, biological productivity, and carbon sequestration [1]. The timing and location of this meltwater release are key factors affecting both regional and large-scale oceanic circulation. Therefore, accurately modeling and understanding the dynamics of iceberg melt is essential for predicting its impact on oceanic circulation, nutrient release, and climate feedback mechanisms [3, 4].

Interest in iceberg melting grew in the 1970s and 1980s when researchers explored the potential of harvesting icebergs for freshwater [2]. Although practical challenges ultimately led to the abandonment of these efforts, the studies conducted during this period underscored the complexity of accurately predicting meltwater behavior. Modeling iceberg melting in the laboratory also has its challenges, such as the difficulty of accessing cold rooms and the complications of taking measurements at freezing temperatures. Recent advancements have renewed interest in understanding iceberg melt, yet traditional experimental approaches remain limited. Given the difficulties in observing small-scale turbulent processes, numerical simulations have become essential for exploring these dynamics and their broader implications for ocean circulation and ecosystems [2].

Melting in binary mixtures, such as ice in seawater, is traditionally modeled as a moving boundary problem, where separate partial differential equations are applied in the liquid and solid regions with boundary conditions at the interface. This method, while effective, poses significant challenges due to the dynamic nature of the boundary, complicating numerical simulations and mathematical

proofs [2]. To address these complexities, phase-field models offer a promising alternative by representing distinct phases within the system using a single, smoothed phase field ϕ . This approach eliminates the need for explicit boundary tracking, simplifying numerical simulations.

In phase-field models, the interface between phases is represented implicitly by the level set $\phi = 1/2$, where $\phi = 1$ is the solid phase and $\phi = 0$ is the fluid phase. The interface's width, defined by the Cahn number ϵ , is the ratio of the microscopic interface width to the bulk system size. Although phase-field models require resolving the diffuse interface, which may pose challenges in large-scale simulations, their physical motivation and numerical simplicity make them an attractive option for modeling complex multi-phase systems [5].

The phase field ϕ evolves according to an Allen–Cahn type equation, which has been adapted to include the effects of temperature T and concentration C :

$$\frac{5}{6} \frac{\Lambda}{\epsilon c_p \kappa} \partial_t \phi - \gamma \nabla^2 \phi + \frac{1}{\epsilon^2} \phi(1 - \phi) [\gamma(1 - 2\phi) + \epsilon(T + \lambda C)] = 0, \quad (1)$$

where Λ is the latent heat, γ is a constant related to the interface energy, κ is the thermal diffusivity, c_p is the specific heat, and λ represents the coupling between temperature and concentration. This equation describes how the phase field ϕ changes over time, accounting for both diffusive effects and the influence of the temperature and concentration fields.

The governing equations extend the classical Navier-Stokes framework by incorporating phase-field variables, which model the phase transition between solid and fluid phases. This modified system ensures incompressibility, temperature evolution, concentration dynamics, and momentum conservation within the context of phase-field theory.

The incompressibility condition is given by:

$$\nabla \cdot \vec{u} = 0, \quad (2)$$

where $\vec{u}$ represents the fluid velocity field.

The temperature field T evolves as:

$$\partial_t T + \nabla \cdot [(1 - \phi) \vec{u} T - \kappa \nabla T] = \frac{\Lambda}{c_p} \partial_t \phi, \quad (3)$$

Similarly, the Salinity field C is governed by:

$$\partial_t [(1 - \phi + \delta) C] + \nabla \cdot [(1 - \phi + \delta)(\vec{u} C - \mu \nabla C)] = 0, \quad (4)$$

Finally, the momentum equation, incorporating phase-field effects, is:

$$\partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} - \nu \nabla^2 \vec{u} + \nabla p + \frac{g\rho(T, C)}{\rho_0} \hat{z} = -\frac{\nu}{\eta} \phi \vec{u}, \quad (5)$$

where ν is the kinematic viscosity, p is the pressure, and the buoyancy force $\frac{g\rho(T, C)}{\rho_0} \hat{z}$ accounts for temperature and concentration variations.

These equations collectively extend the classical Navier-Stokes equations by integrating the dynamics of phase transitions, capturing the complex interactions between thermal, solutal, and mechanical forces in multi-component flows [5].

There have been numerous studies investigating the dependence of iceberg melt rates on various environmental factors. The influence of ambient water temperature has been explored extensively [1], with additional research focusing on the impact of salinity [2] and aspect ratio [6] on melting dynamics. However, the impact of the Coriolis force on iceberg melting has often been neglected.

The Coriolis force is expected to influence heat transfer by altering flow patterns. The Coriolis force is expected to influence heat transfer by altering flow patterns [7]. Therefore, in this project, I aim to explore the combined impact of rotation and salinity. To incorporate the effects of the Coriolis force, I modified the momentum equation as follows:

$$\partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} - \nu \nabla^2 \vec{u} + \nabla p + \frac{g\rho(T, C)}{\rho_0} \hat{z} + 2\Omega \sin \phi \hat{z} \times \vec{u} = -\frac{\nu}{\eta} \phi \vec{u}, \quad (6)$$

where Ω represents the angular velocity of rotation and ϕ the latitude. Equations (1) to (4) are used as originally presented in the study.

Method

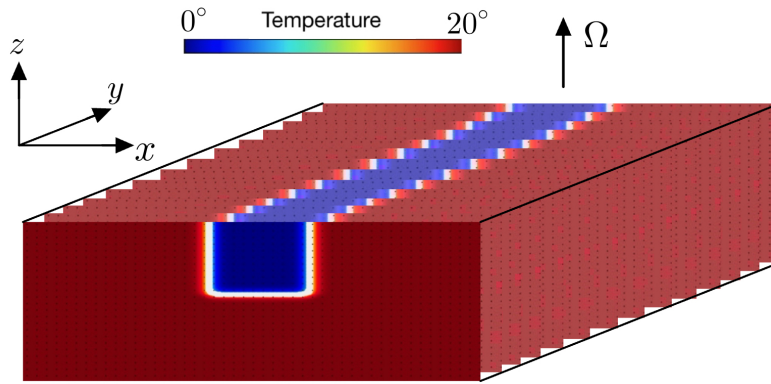


Figure 1: **3D representation of the simulation setup, with the y -direction extending to infinity.** The system consists of a rotating rectangular tank with a square ice sheet placed at the top center of the $x - z$ plane. Melting occurs identically along each $x - z$ cross-section as the system rotates around an axis parallel to the z -axis, passing through the ice center.

In this study, I explored the impact of rotation and salinity on ice melting through two-dimensional numerical simulations conducted using the Python code "Dedalus 3". The simulation code, originally developed in Dedalus 2 to solve Equations (1) to (5), was based on the work "Aspect ratio affects iceberg melting" by Hester et al. (2021). I modified the code to run in Dedalus 3 and to incorporate the influence of the Coriolis force on the fluid dynamics by replacing Equation 5 with Equation 6 and adjusting the code accordingly. Furthermore, I replaced the SinCos basis with the Chebyshev basis in the vertical (z) axis to more accurately enforce the boundary conditions, utilizing generalized Tau methods [8].

The system consists of a rotating rectangular tank, initially measuring 10 cm in width and 5 cm in height, with the y direction extending to infinity. A 2 by 2 cm ice sheet at 0°C and zero salinity is placed at the top center of the tank, which is filled with 20°C water at varying levels of salinity for each trial. The tank, positioned at a latitude of 10°, can rotate at different angular velocities around an axis parallel to the z -axis, passing through the center, driven by an external force.

As shown in Figure 1, the 3D representation of the system highlights that each $x - z$ slice melts identically due to the 2D nature of the simulation. Since this is a 2D simulation, each $x - z$ slice melts identically, and the partial derivative of any physical quantity (such as flow or temperature) with respect to y is zero. Due to the computational expense of the simulation, particularly at salinity levels above 20 $\frac{\text{g}}{\text{kg}}$, I rescaled the setup to a 5 by 2.5 cm $x - z$ cross-section containing a 1 by 1 cm ice sheet to explore the impact of salinity.

The simulations were conducted using a grid resolution of 512 points in the horizontal direction and 256 points in the vertical direction. The fluid dynamics were governed by a Prandtl number of 7 and a Schmidt number of 12.5, with the kinematic viscosity set to $1.3 \times 10^{-2} \text{ cm}^2/\text{s}$. Material properties included a specific heat capacity of 4.2 J/g°C and a latent heat of fusion of $3.34 \times 10^2 \text{ J/g}$, with the phase field interface thickness (ϵ) set to 0.04 cm in both setups due to the instability of the simulation. Additionally, I investigated the impact of salinity across Schmidt numbers 14 and 20, with salinity levels ranging from 2 to 20 $\frac{\text{g}}{\text{kg}}$.

For all simulations, the boundary conditions were impermeable and stress-free for velocity, with Neumann conditions for temperature, concentration, and phase field at both the top and bottom boundaries ($z = 0$ and $z = \text{height}$). A pressure gauge condition was enforced to maintain pressure consistency, and periodic boundary conditions were applied in the horizontal direction (x).

The simulations were executed using the computing resources provided by the Digital Research Alliance of Canada [9].

Results and Discussion

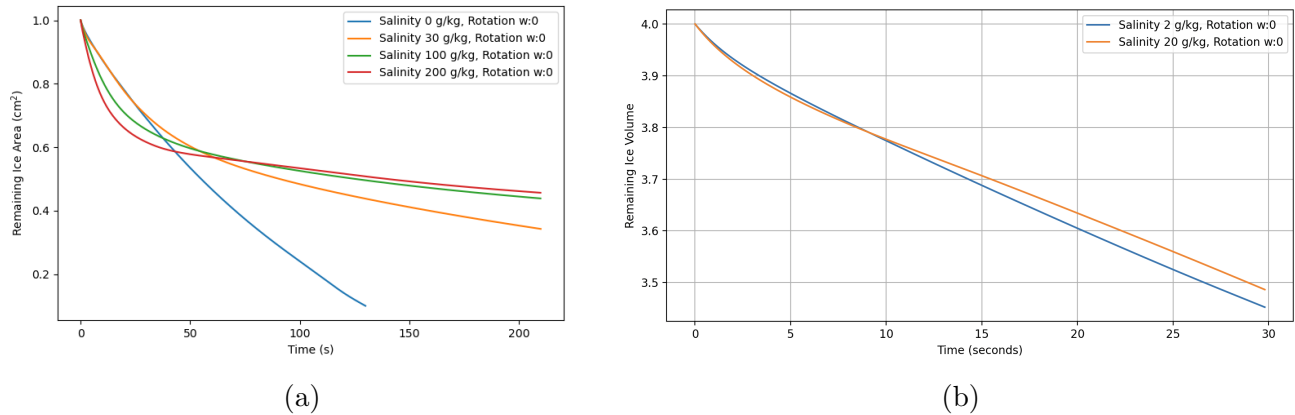


Figure 2: **Effect of Salinity on Ice Melting Rates in a Non-Rotating Environment** (a) Time series showing the remaining ice area (cm²) at different salinity levels (0 g/kg, 30 g/kg, 100 g/kg, and 200 g/kg). Initially, higher salinity accelerates the melting rate due to freezing point depression. Over time, increased salinity stabilizes thermal gradients, resulting in slower melting. The time to reach 50% of the initial ice volume is shortest at 0 g/kg and increases with salinity. (b) Similar to (a), this graph shows the effect of salinity at 2 g/kg and 20 g/kg, highlighting an earlier onset of the negative impact on the melting rate at lower salinity levels.

Figures 2a and 2b depict the remaining ice volume, expressed in cm², over time at various salinity levels. To assess the effect of salinity on the ice melting rate, it is advisable to prolong the simulation duration to reduce the influence of other potential unknown factors, as outlined in Appendix A. Initially, an increase in salinity accelerates the ice melting rate, likely due to the reduction in the freezing point. This decrease causes the surrounding water to freeze more slowly, shifting the equilibrium between water and ice towards melting. However, as time progresses, the salinity begins to exert a negative impact on the melting rate. This inverse effect may be attributed to the fact that higher salinity levels increase the density of the water near the ice, which stabilizes the stratification. This stabilization reduces the mixing of warmer water from below, leading to a decrease in the heat transfer to the ice surface and thus slowing down the melting process. Therefore, as observed in Figure 2a, the time required for the ice volume to decrease to 50% of its initial value is shortest at 0 g/kg salinity, followed by 30 g/kg, then 100 g/kg, and finally longest at 200 g/kg salinity. Similarly, the same pattern is evident in Figure 2b, where the negative impact of salinity on the ice melting rate is observed earlier at lower salinity levels, possibly due to the less pronounced freezing point depression, which causes a quicker stabilization of the thermal gradients.

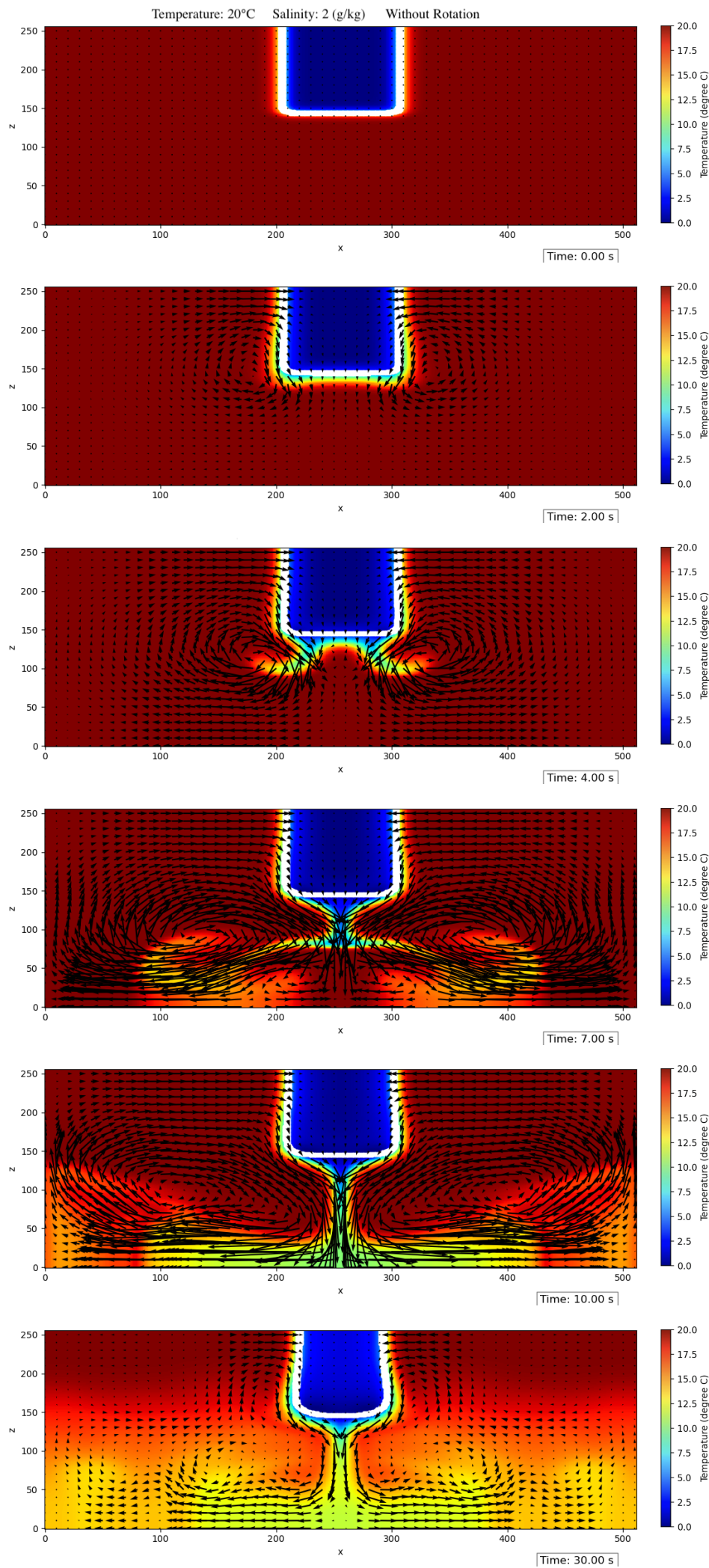


Figure 3: Illustrates the ice melting process in 2 g/kg salinity at 20°C, showing the evolution of temperature fields and velocity vectors as convective flows develop and accelerate the melting over time.

Figure 3 presents a sequence of images depicting the ice melting process in a water-filled container with a salinity of 2 g/kg. At time 0 s, the system is in its initial state, with the ice at 0°C and the surrounding water at a uniform temperature of 20°C. At this stage, the ice has not yet begun to melt, and the white line around the ice represents the phase field interface (ϵ) between water and ice.

By 2 seconds, heat transfer from the warmer water to the ice initiates the melting process. The ice melts most rapidly at the sharp edges, where the increased surface area allows more water to contact and heat the ice. As a result, the sharp edges of the ice begin to round over time. Warm water rises, displacing the cooler, denser water downward, which initiates vertical convective flow around the ice and leads to the formation of small circular patterns in the water.

At 4 seconds, the flow patterns become more pronounced as warm water rises and cooler water sinks, reinforcing the circulation that enhances the melting process. Distinct vortices form below the ice, driven by the fluid's attempt to redistribute heat and maintain equilibrium. With no rotation, the Coriolis effect is absent, and at a salinity of only 2 g/kg, these vortices are primarily driven by thermal convection. The low salinity means that the density differences are mainly due to temperature variations.

By 7 seconds, the vortex structures become more visible, and the melting of the ice is more evident at the bottom and sides, where heat transfer is most efficient. The vortices enhance mixing, continuously bringing warmer water into contact with the ice, thereby accelerating the melting rate.

By 10 seconds, the temperature gradient around the ice remains sharp, with cooler water still surrounding the ice. The melting has progressed significantly, with the ice block showing signs of reduced size, particularly at the bottom where warm water continually rises and contacts the ice. Consequently, the temperature at the bottom of the container has significantly dropped.

At 30 seconds, the temperature in the container begins to homogenize due to the mixing induced by convective currents, but complete homogenization has not yet been achieved. Vortex structures are still evolving, indicating that the system has not yet reached a steady state. The flow continues to adjust, and temperature gradients continue to drive changes in the circulation patterns. The melting process remains active, and the system is still far from equilibrium.

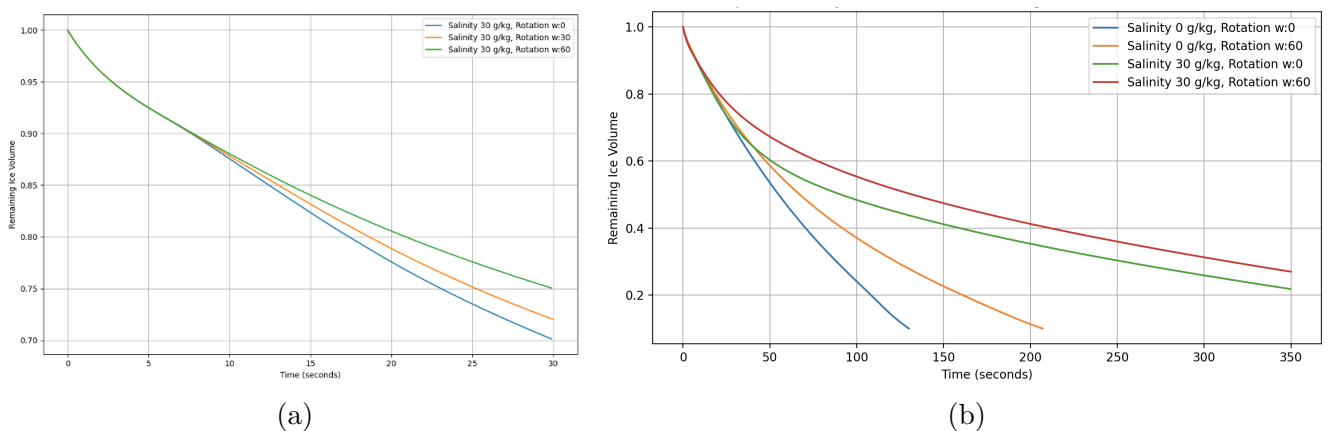


Figure 4: **The impact of rotation on ice melting rate at 0 and 30 g/kg salinity.** In (a), increased angular velocity reduces the melting rate due to suppressed vertical convection. In (b), the inverse effect of rotation on melting rate increases over time at 0 g/kg, but it stabilizes at 30 g/kg after reaching stratification.

As shown in Figure 4a, at 30 g/kg salinity, increasing the angular velocity decreases the ice melting rate. This reduction can be attributed to the Coriolis effect, which deflects fluid motion and suppresses vertical convection, leading to less efficient heat transfer between the water and the ice. Consequently, the melting process slows down.

In contrast, Figure 4b demonstrates that at zero salinity, the impact of rotation on ice melting continues to increase over time, showing a consistent divergence between different rotational speeds. However, at 30 g/kg salinity, the initial difference stabilizes, and the effect of rotation no longer diverges further. This behavior could be due to the stabilization of stratification caused by higher salinity, which reduces the system's sensitivity to rotational effects after an initial adjustment period.

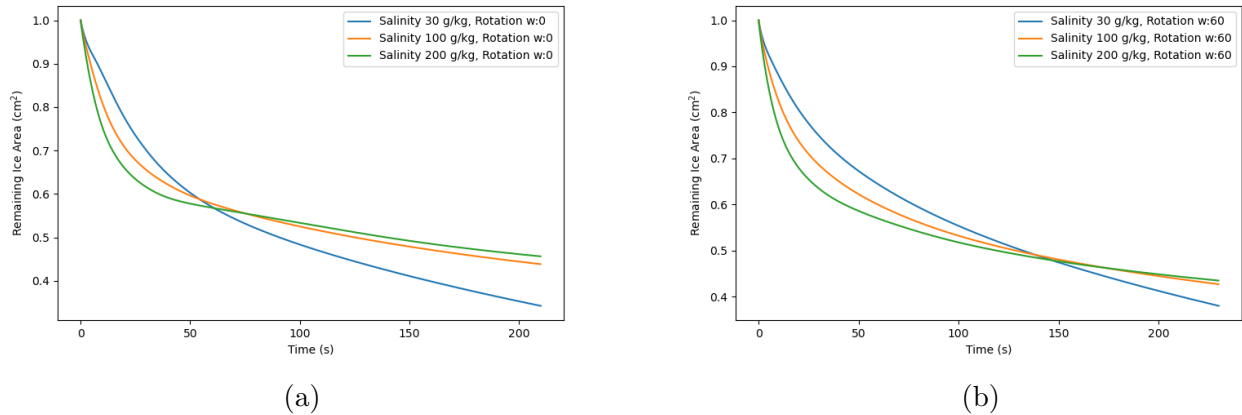


Figure 5: **Comparison of ice melting rates at different salinities with and without rotation.** Initially, higher salinity accelerates melting, but stratification slows the process around 55 seconds in (a). In (b), with rotation at 60 rpm, the stabilizing effect of rotation delays the impact of salinity, causing stratification to dominate much later, around 150 seconds.

In plot 5a, where the angular velocity is zero, it is observed that initially, ice melts faster at a salinity of 100 g/kg compared to 30 g/kg. However, around 55 seconds, the effect of stratification becomes dominant, reversing the trend and causing the ice in the 30 g/kg salinity water to melt faster. A similar pattern is observed when comparing salinities of 200 g/kg and 30 g/kg, though the difference in melting rates is more pronounced, with the inverse effect of salinity on ice melting occurring later, around 65 seconds.

In contrast, plot 5b, where the angular velocity is 60 rpm, shows that the negative impact of higher salinity becomes dominant much later—around 150 seconds for both 100 g/kg and 200 g/kg salinities compared to 30 g/kg. This delay is likely due to the stabilizing effect of rotation, which suppresses vertical convection and stratification, thereby reducing the immediate impact of salinity differences.

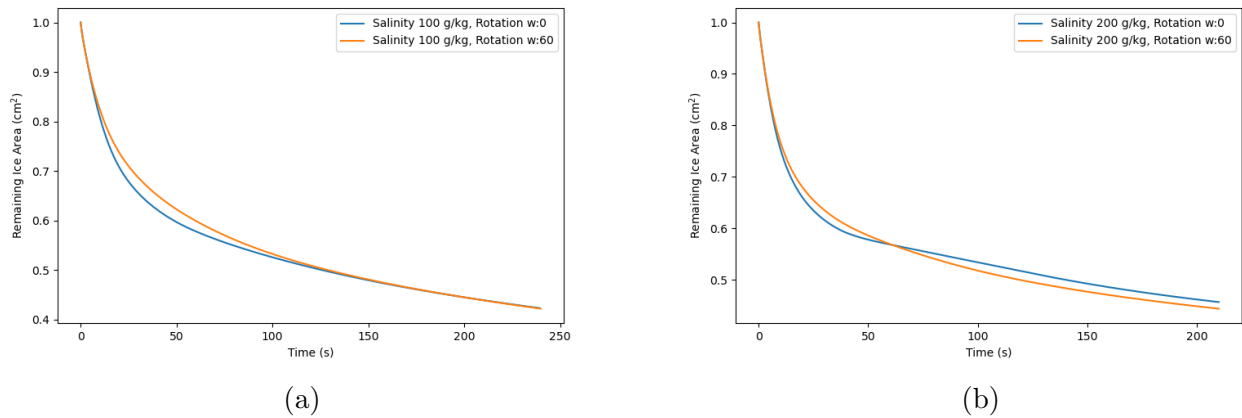


Figure 6: **The impact of rotation on ice melting rate at high salinity levels.** Initially, the non-rotating setup exhibits a higher melting rate due to more direct vertical heat transfer. However, as the system stabilizes, rotation enhances the melting process due to thermal wind balance

Figure 6 illustrates the remaining ice volume over time at salinity levels of 100 g/kg and 200 g/kg, comparing cases with and without rotation. Initially, in both scenarios, the non-rotating setup exhibits a higher melting rate, as indicated by the lower remaining ice volume. This occurs because the absence of rotation allows for more direct vertical heat transfer from the surrounding water to the ice, leading to faster melting. However, as the system evolves and reaches stable stratification, the presence of rotation begins to enhance the melting process. To further understand the influence of rotation on the melting process, I examined the temperature fields and flow patterns observed at various time points, as shown in Figure 7.

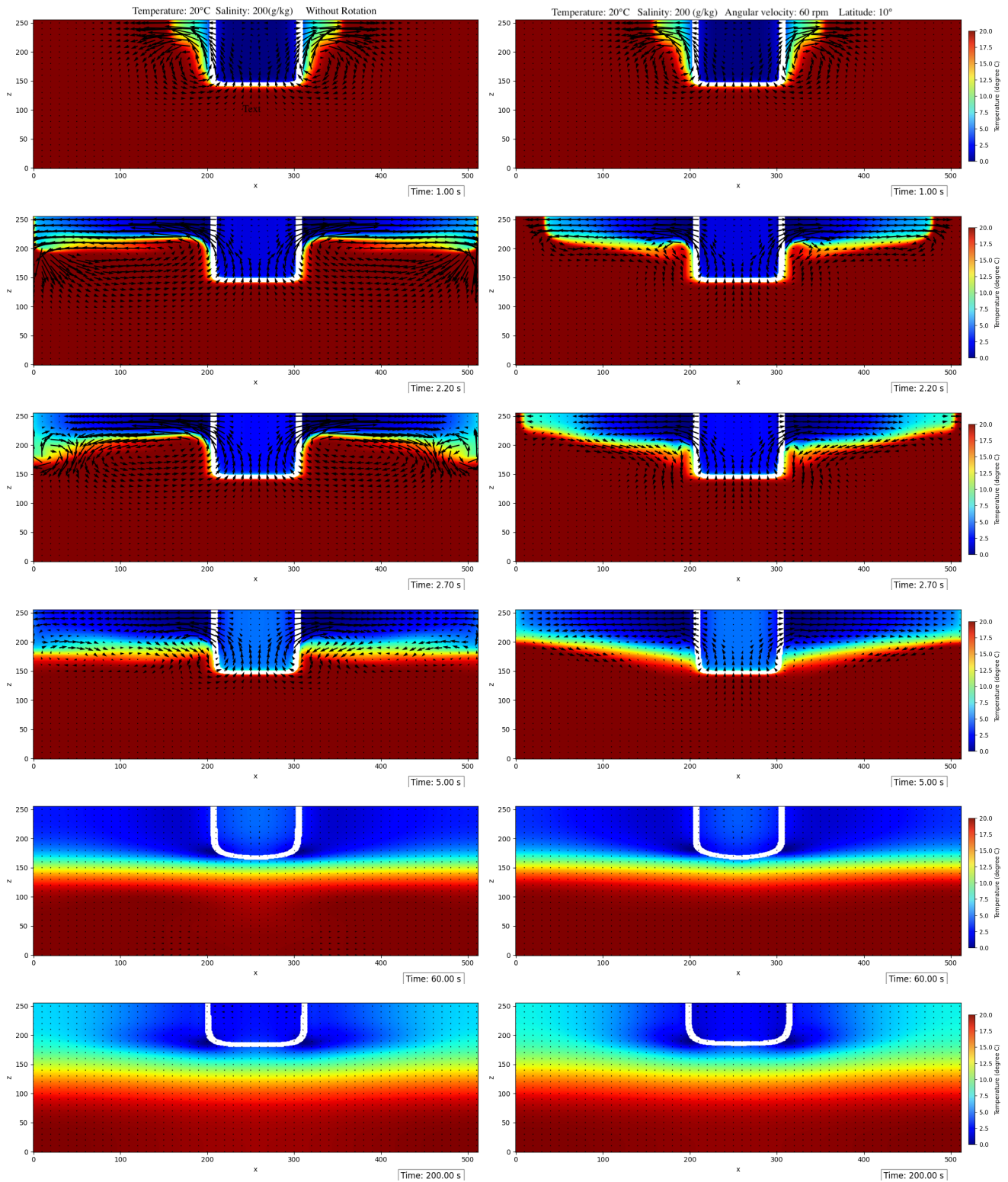


Figure 7: **Influence of Rotation on Ice Melting at 200 g/kg Salinity.** For the first 60 seconds, the ice melting rate is higher in the non-rotating case due to more efficient vertical heat transfer. As the stratification stabilizes, rotational effects begin to enhance the melting process through thermal wind balance.

Figure 7 illustrates the evolution of flow dynamics and temperature distribution, gradually diverging between the rotating and non-rotating cases. At 1.00 seconds into the simulation, both scenarios display nearly identical flow patterns and temperature gradients, indicating that rotational effects have not yet introduced significant differences at this early stage. This is because the period of rotation is 1 second, which is too brief for these effects to fully manifest, leaving the flow primarily governed by initial conditions. In both cases, colder meltwater rises along the sides of the ice, while warmer, salty water is displaced downward, forming vortices around the ice. The high salinity plays a dominant role, making the fresh meltwater much lighter than the surrounding salty warm water, which results in a strong buoyancy force that pushes the meltwater to the surface and generates swirling flows around the ice cube.

By 2.20 seconds, the impact of rotation becomes evident. In the non-rotating scenario, the

cold meltwater rapidly reaches the wall, displacing the warmer water downward. In contrast, the rotating system experiences the Coriolis force deflecting the upward flow of cold water diagonally, delaying its contact with the wall. Near the ice boundary, the balance between Coriolis and viscous forces causes the fluid to spiral, forming an Ekman layer—a characteristic pattern of rotating systems.

At 2.70 seconds, in the non-rotating case, the cold meltwater that has reached the wall creates a return flow, producing a secondary circulation pattern that pushes the water back toward the ice. Meanwhile, in the rotating case, the two diagonal lines visible in the temperature field are due to circular fluid motions driven by the dominant salinity gradient and the Coriolis effect. These circular motions create a spiraling pattern that manifests as diagonal lines in the cross-sectional view.

By 5.00 seconds, the non-rotating case exhibits more turbulence, with larger velocity vectors compared to the rotating case. In the non-rotating scenario, the side surfaces of the ice remain in contact with the warm water, whereas in the rotating case, only the bottom surface of the ice is in contact with the warmer water. This difference is attributed to the deviation of flow in the rotating case, where the Coriolis effect causes the fresh meltwater to shift away from the ice. In contrast, in the non-rotating case, the cold fresh water rises more vertically, leading to faster melting as the warm water directly contacts more of the ice's surface.

After one minute, approximately when the remaining ice volume curves in Figure 7b intersect, both setups exhibit clear stratification with distinct temperature layers, primarily driven by high salinity. Subtle differences in heat distribution remain: the dark blue layer of cold water is slightly thinner in the rotating case, particularly near the walls, suggesting that the warmer water layer, ranging from dark red to green, is somewhat thicker than in the non-rotating scenario. Additionally, in the rotating case, the hottest temperature layer near the bottom is more uniform, while in the non-rotating case, there is a mild brightening in the corresponding layer below the ice. These minor differences will become more pronounced, leading to significant variations in temperature distribution as the simulation progresses to 200 seconds.

By 200 seconds, the differences in heat distribution have become more pronounced. In the rotating case, the top right corner of the figure shows a higher temperature, indicated by green, compared to the blue seen in the non-rotating case. This suggests that in the rotating system, the ice is surrounded by warmer water as one moves horizontally from the ice surface toward the container walls. The observed difference in melting rates between the two setups from 60 to 200 seconds can be explained as follows: around 60 seconds, the ice melting rate in both setups significantly decreased, leading to more stable stratification. As melting continues, the small volume of fresh, cold meltwater generated is insufficient to disrupt this stratification, resulting in slow water movement. However, in the rotating system, as fluid moves horizontally away from the center, its tangential velocity increases to conserve angular momentum due to the constant angular velocity of 60 rpm. This flow achieves a thermal wind balance, where the Coriolis force and the horizontal pressure gradient due to temperature differences work together to create a steady, balanced flow. This balance organizes the horizontal temperature gradients more effectively than in the non-rotating case, enhancing the horizontal transport of warmer water. As a result, the structured horizontal flow in the rotating system contributes to a slightly accelerated melting rate compared to the non-rotating system.

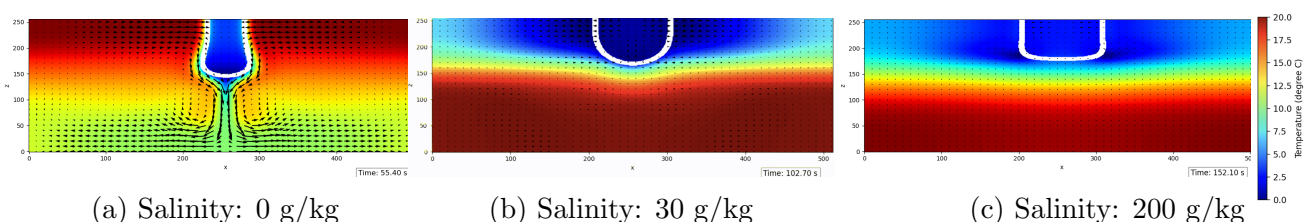


Figure 8: **Comparison of the ice geometry at salinity levels of 0 g/kg, 30 g/kg, and 200 g/kg in the surrounding water when the remaining ice volume reaches 50% of its initial size.** The figure demonstrates that ice surrounded by higher salinity water has a smaller surface area at a specific volume.

Finally, I observed that the geometry of the remaining ice varies significantly at different salinity

levels (0, 30, and 200 g/kg) when the ice volume reaches 50% of its initial size. Specifically, at lower salinity levels, the ice retains a larger surface area at the same volume compared to higher salinity levels, as illustrated in Figure 8. This variation in surface area can be attributed to differences in the melting patterns of the ice at different salinity levels. At zero salinity, the ice initially melts at the bottom corners due to their larger surface area compared to other parts. As the cold meltwater sinks, it displaces the warmer water upward, causing the ice's vertical surfaces become concave and increasing the overall surface area. Likewise, at 30 g/kg salinity, melting starts at the bottom corners, rounding these areas. A similar flow pattern is initially observed, leading to melting along the vertical sides. However, as stratification occurs, melting becomes confined to the lower part of the ice. At 200 g/kg salinity, stratification happens more rapidly, causing the ice to melt primarily from its bottom surface, resulting in a less rounded, more rectangular remaining ice volume.

Since the rate of melting increases with surface area, this finding may offer another possible explanation for why I observed a larger remaining ice volume at 200 g/kg salinity compared to 30 g/kg salinity after approximately 65 seconds.

Conclusion

Iceberg melting plays a crucial role in the redistribution of freshwater from ice sheets into the oceans, significantly impacting oceanic circulation and marine ecosystems. In this study, I conducted numerical simulations to investigate the effects of salinity and rotation on ice melting using the Dedalus 3 framework.

I observed, Initially, higher salinity levels accelerate the melting process, likely due to melting point depression. However, over time, increased salinity stabilizes thermal gradients, reducing the melting rate. Furthermore, the amount of salinity in the water influences the geometry of the remaining ice, such that at the same remaining volume, higher salinity results in a smaller surface area. This reduction may further explain the decrease in the melting rate observed as salinity increases after the initial phase.

The impact of rotation on ice melting is more pronounced at higher angular velocities and is closely related to the salinity of the surrounding water. At zero salinity, rotation decreases the ice melting rate over time by suppressing vertical convection. At 30 g/kg salinity, the melting rate initially decreases due to rotation, but once stratification occurs, the impact of rotation on the melting rate becomes insignificant. In contrast, at higher salinity levels, such as 100 g/kg and 200 g/kg, rotation initially reduces the melting rate, but after near-stable stratification is reached, rotation actually increases the rate, possibly due to the development of thermal wind balance.

For future work, it is recommended to examine a broader range of salinity levels and angular velocities. Additionally, the simulations in this study only considered a water temperature of 20°C; therefore, evaluating the impact of salinity and rotation at different temperatures would provide a more comprehensive understanding. A potential next step could be to incorporate salinity into the ice itself, as the current simulations assumed the ice had zero salinity. Comparing the time required for the ice to reach near-zero volume in each trial could yield more reliable insights. Moreover, using more realistic non-dimensionalized parameters, such as the Schmidt number and Reynolds number, would enhance the reliability and applicability of the results.

Acknowledgments

I would like to express my sincere gratitude to the Natural Sciences and Engineering Research Council of Canada (NSERC) for awarding me the Undergraduate Student Research Award (USRA), which made this research possible. I am also deeply thankful to my supervisors, Prof. Nicolas Grisouard and Prof. Erica Rosenblum, for their invaluable guidance and support throughout this project.

References

- [1] E. W. Hester, C. D. McConnochie, C. Cenedese, L.-A. Couston, and G. Vasil, “Aspect ratio affects iceberg melting,” *School of Mathematics and Statistics, University of Sydney*, Sydney, NSW 2006, Australia, 2021.
- [2] C. Cenedese and F. Straneo, “Icebergs Melting,” *Annual Review of Fluid Mechanics*, vol. 55, pp. 377-402, 2023.
- [3] Tournadre et al., “Iceberg trajectories and their impact on ocean currents and climate,” *Mar. Geophys. Res.*, 2015.
- [4] Bigg, “Impact of iceberg melting on sea ice formation and albedo,” *Global Climate Change*, 2016.
- [5] E. W. Hester, L.-A. Couston, B. Favier, K. J. Burns, and G. M. Vasil, “Improved phase-field models of melting and dissolution in multi-component flows,” *Proc. R. Soc. A*, vol. 476, 2020.
- [6] FitzMaurice and Stern, “Aspect ratio effects on iceberg melting,” *Geophysical Research Letters*, 2018.
- [7] A. N. Meroni, C. D. McConnochie, C. Cenedese, K. Snow, “Nonlinear influence of the Earth’s rotation on iceberg melting,” *Journal of Fluid Mechanics*, vol. 858, pp. 832-851, 2019.
- [8] Dedalus Project. *Tau Method Documentation*. Available at: https://dedalus-project.readthedocs.io/en/latest/pages/tau_method.html.
- [9] Digital Research Alliance of Canada. Available at: <https://alliancecan.ca/en>.

Appendix A

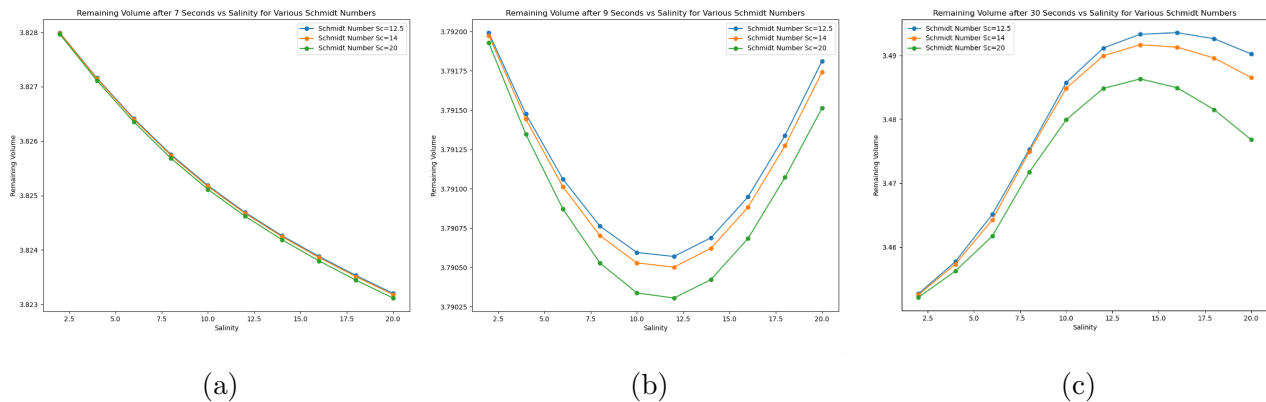


Figure 9: **Remaining ice volume at different salinity levels (2 to 20 g/kg) and 20°C, measured at (a) 7 seconds, (b) 9 seconds, and (c) 30 seconds after the simulation begins.** The impact of salinity on ice melting varies with observation time, highlighting the need for extended simulations or comparisons at specific time points. Additionally, the effect of salinity is influenced by the Schmidt number, as shown in plot (c), where increasing salinity from 14 g/kg to 16 g/kg reduces the melting rate at a Schmidt number of 12.5, but increases it at a Schmidt number of 20. This underscores the importance of carefully selecting simulation parameters.

Figure 9 depicts the remaining ice volume at various salinity levels, ranging from 2 to 20 g/kg, at a temperature of 20°C, measured at 7 seconds, 9 seconds, and 30 seconds after the simulation begins. The results demonstrate that conclusions about the impact of salinity on the ice melting rate can vary significantly depending on the time at which the simulation results are observed. This indicates that to accurately assess the influence of a variable such as salinity, it is essential to either extend the simulation over a longer period to observe the development of its effects on the melting rate or to compare the time required for the ice to reach a specific percentage of its initial volume. By doing so, we can minimize the impact of other potential unknown variables on the interpretation of the results.

Another important consideration is how different outcomes can be obtained based on the selected parameter values. For instance, in Figure 9c, at a Schmidt number of 12.5, increasing salinity from 14 g/kg to 16 g/kg decreases the melting rate, whereas at a Schmidt number of 20, the effect is the opposite. Since the Schmidt number is the ratio of momentum diffusivity (kinematic viscosity) to mass diffusivity (salinity diffusion in this context), the choice of the Schmidt number should strike a balance between the computational cost of the simulation and the significance of the variable under observation.

Appendix B

The following information is sourced from the Digital Research Alliance of Canada [9].

Installing Dedalus in a Python Virtual Environment

1. Load Dedalus runtime dependencies.

```
[name@server ~]\$ module load StdEnv/2023 gcc openmpi mpi4py/3.1.4 fftw-mpi/3.3.10
hdf5-mpi/1.14.2 python/3.11
```

2. Create and activate a Python virtual environment.

```
[name@server ~]\$ virtualenv --no-download ~/dedalus_env
[name@server ~]\$ source ~/dedalus_env/bin/activate
```

3. Install a specific version of Dedalus and its Python dependencies.

```
(dedalus_env) [name@server ~]\$ pip install --no-index --upgrade pip
(dedalus_env) [name@server ~]\$ pip install --no-index dedalus==X.Y.Z
```

where X.Y.Z is the exact desired version, for instance 3.0.2. You can omit it to specify the version in order to install the latest one available from the wheelhouse.

4. Validate it.

```
(dedalus_env) [name@server ~]\$ python -c 'import dedalus'
```

5. Freeze the environment and requirements set.

```
(dedalus_env) [name@server ~]\$ pip freeze --local > ~/dedalus-3.0.2-requirements.txt
```

6. Remove the local virtual environment.

```
(dedalus_env) [name@server ~]\$ deactivate && rm -r ~/dedalus_env
```

Running Dedalus

1. Write your job submission script.

```
#!/bin/bash
#SBATCH --account= # name of the allocation account provided by the supervisor or the
# institution. This account is different from the personal account)
#SBATCH --time=00:30:00 # wall-time (hh:mm:ss)
#SBATCH --ntasks=32 # number of tasks
#SBATCH --mem-per-cpu=4G # memory per CPU
#SBATCH --output=slurm-%j.out
#SBATCH --error=slurm-%j.err

module --force purge # Not in the official documentation

# Load required modules
module load StdEnv/2023 gcc openmpi mpi4py/3.1.4 fftw-mpi/3.3.10 hdf5-mpi/1.14.2
python/3.11

# Activate virtual environment
source ~/dedalus_env/bin/activate

export OMP_NUM_THREADS=1

# Run your Python script
srun python \${SCRATCH}/file_name.py
```

- `output=slurm-%j.out`: Directs the standard output of the job to a file. The `%j` is replaced by the job ID.
- `error=slurm-%j.err`: Directs the standard error output of the job to a file. The `%j` is replaced by the job ID.

- **SCRATCH:** Environment variable pointing to a fast storage area allocated for running jobs, where temporary files are stored. The amount of storage available is typically large (in the terabyte range), but files are automatically deleted after a certain period (3 months).

Note: Adjust the number of tasks and memory per CPU based on the specific needs of your code. While increasing the number of tasks can accelerate the simulation, it may also lead to longer wait times in the queue.

Maximum Allowed Wall-Time: The maximum wall-time varies by cluster within the Alliance: Beluga, Cedar, and Graham allow up to 7 days (168 hours), while Niagara permits up to 2 days (48 hours).

2. Submit your job to the scheduler.

```
[name@server ~]$ sbatch submit-dedalus.sh
```

- `scontrol show job %j`: Displays detailed information about the job associated with the specific job ID, including status, resource usage, and start time (note that the start time may not always be accurate). The `%j` is replaced by the job ID.